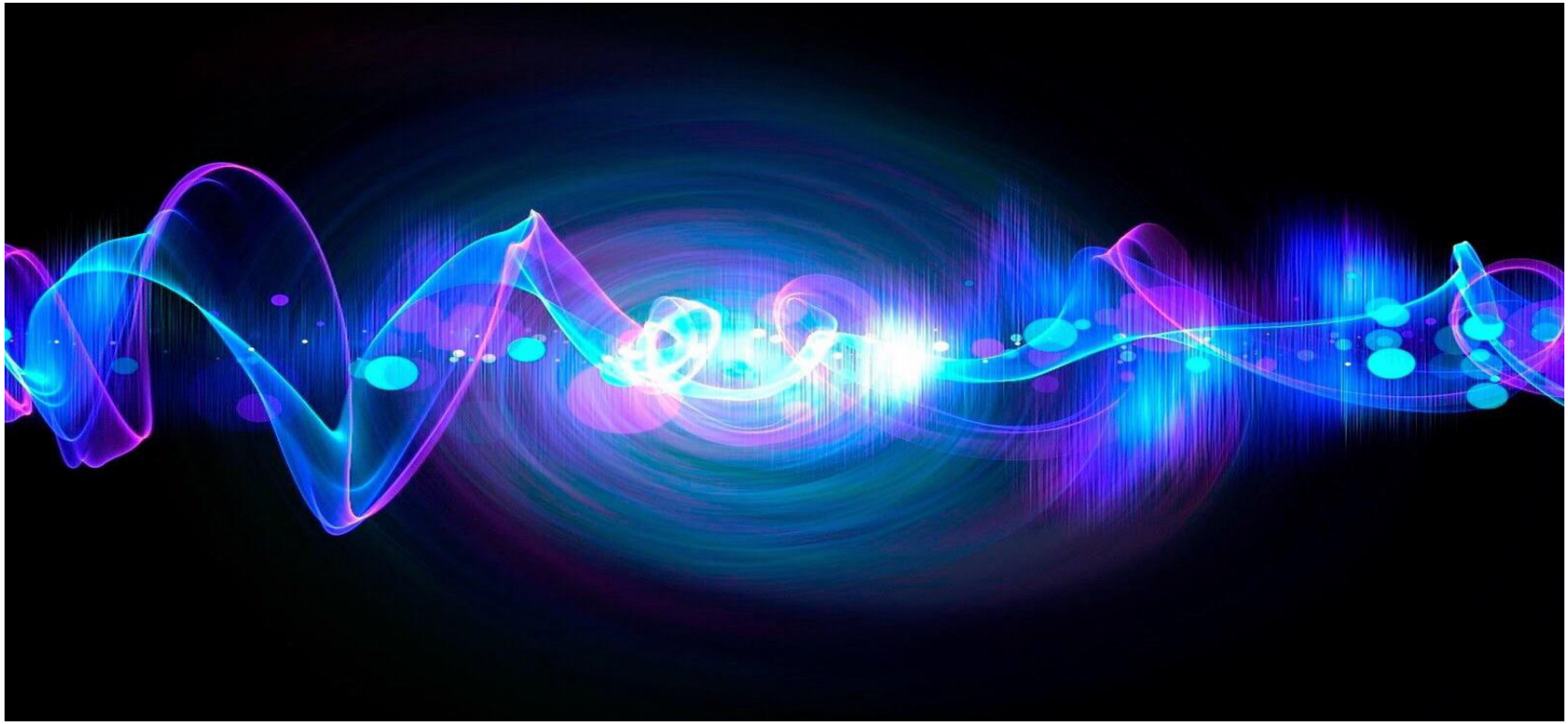
LECTURE 4

Introduction to Waves



What is a wave?

Wave is a vibration or disturbance that travels from one location to another location.

❖ Waves transfers energy without transferring matter.

i. e., carries energy from place to place

❑ A medium:

Is a substance or material that carries the wave

How are waves classified?

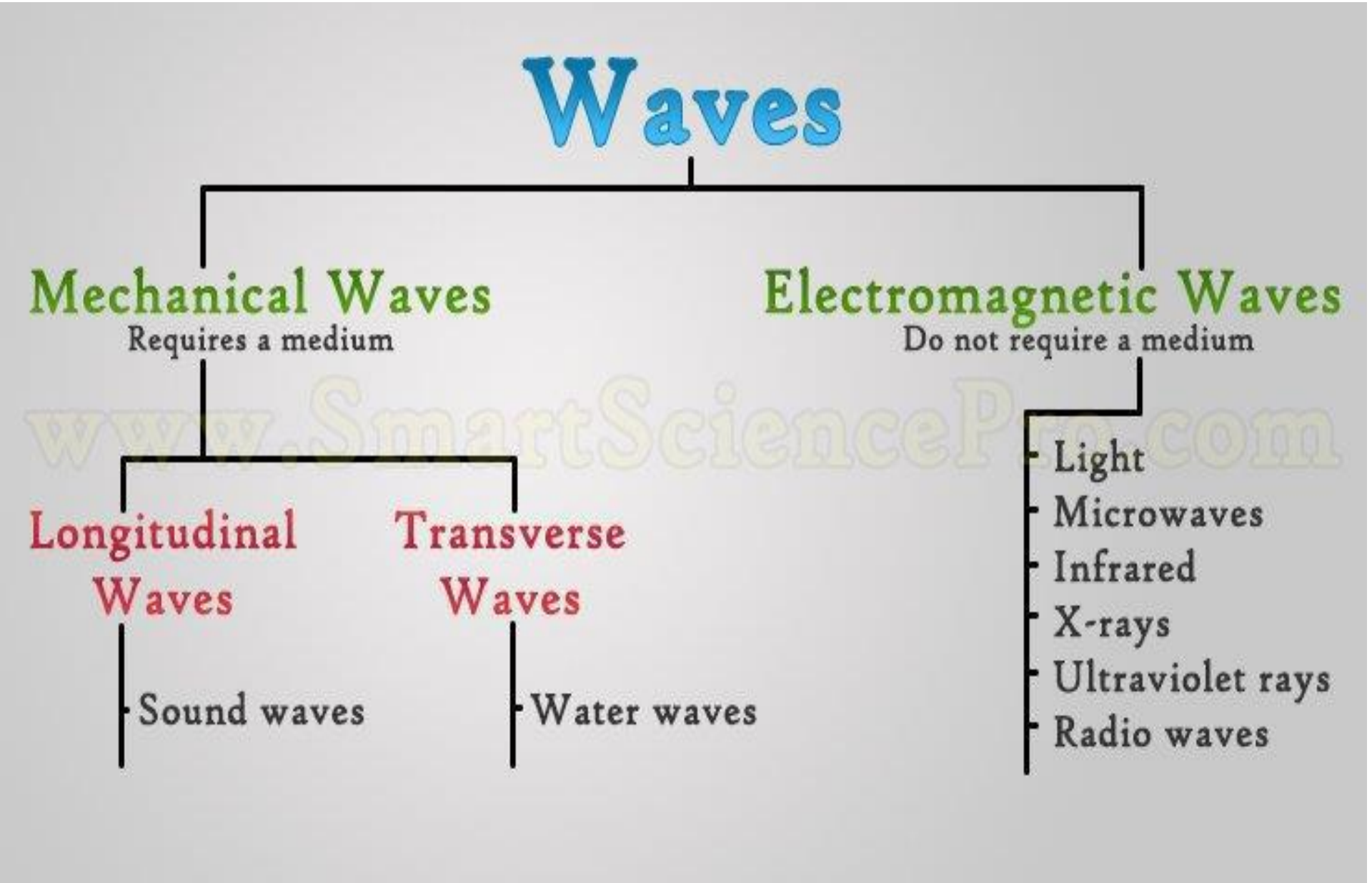
Waves are classified by

WHAT **they move through**

Or by

HOW **particles move through them.**

First : WHAT they move through



1- Mechanical Waves



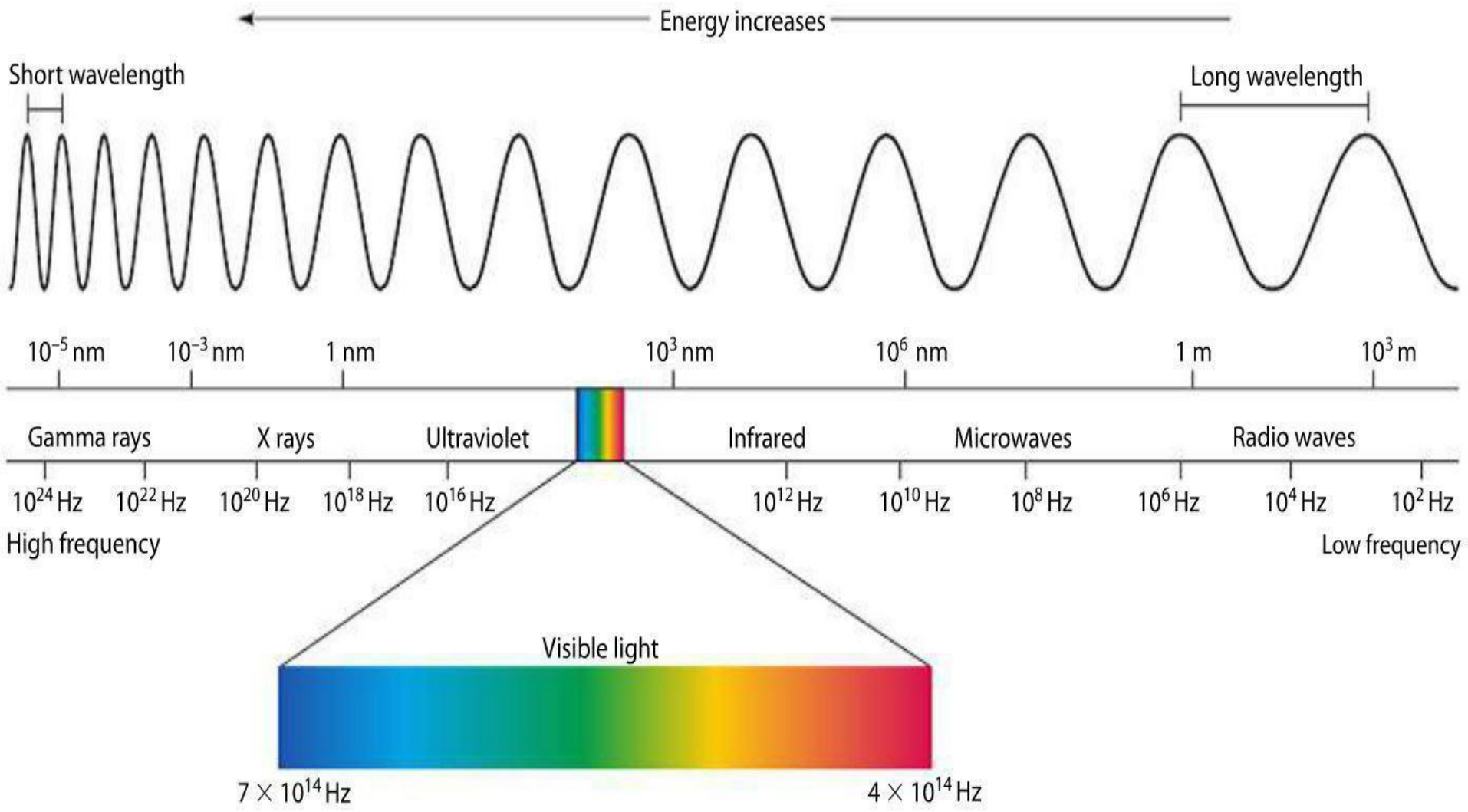
- Needs a medium,
- Require the particles of the medium to vibrate for energy to be transferred.

Example of Mechanical Waves

- water waves,
- earthquake/seismic waves,
- sound waves,
- waves that travel down a rope or spring.

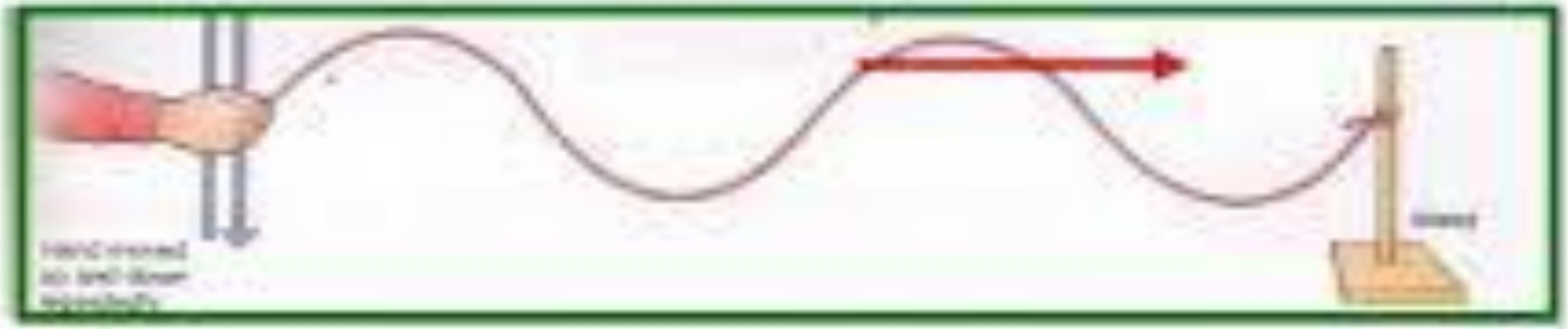
2- Electromagnetic Waves

- **Waves that can travel through matter or empty space (matter is not present).**



Second: HOW particles move through them

- Transverse Waves



- Longitudinal Waves

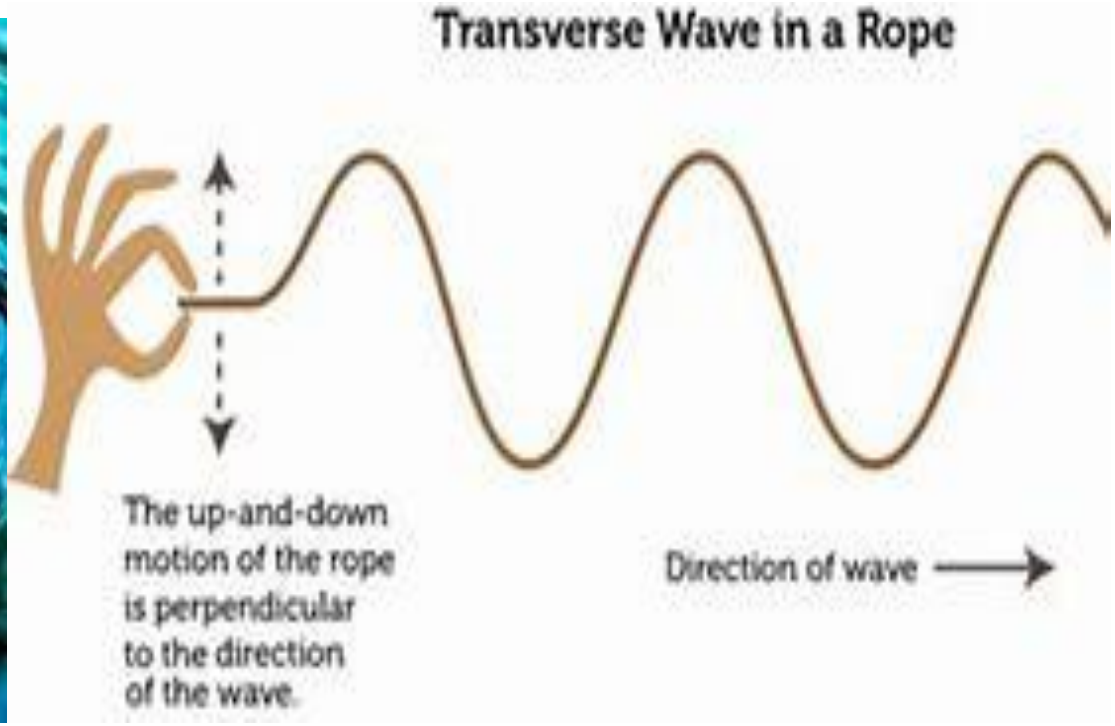


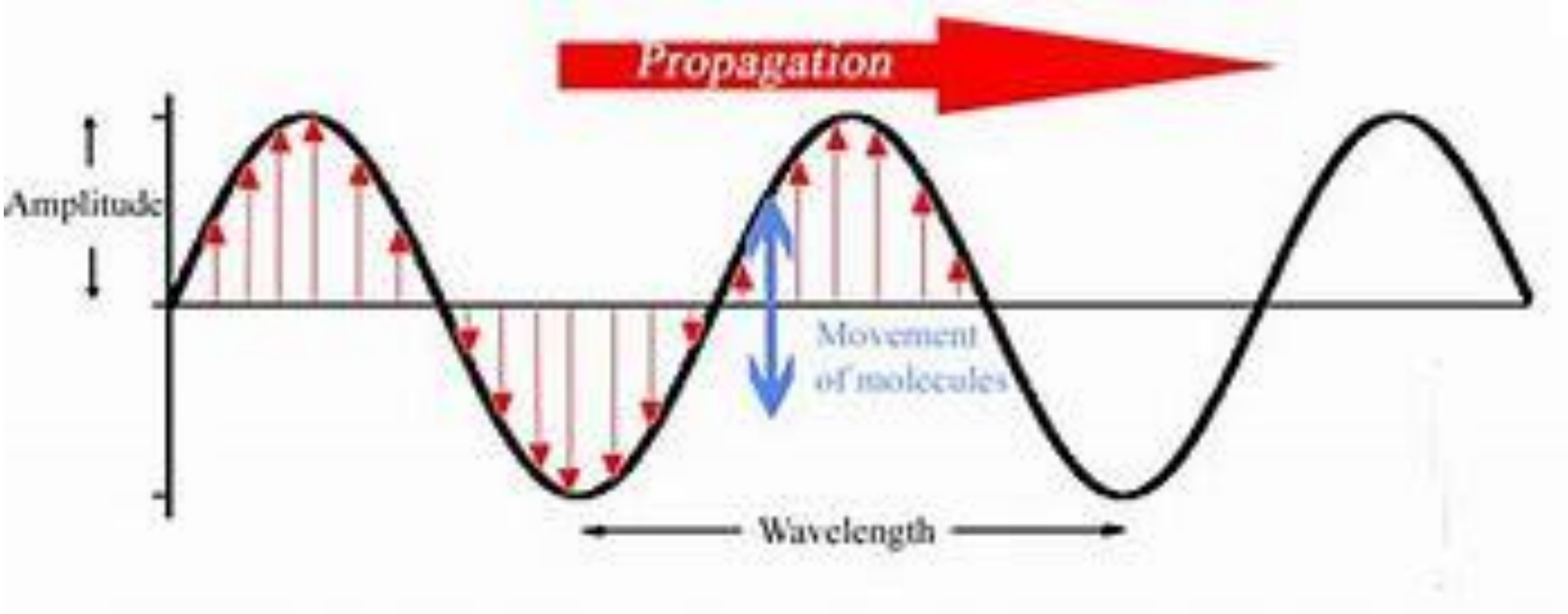
1- Transverse Waves

- Particles move ***perpendicular***

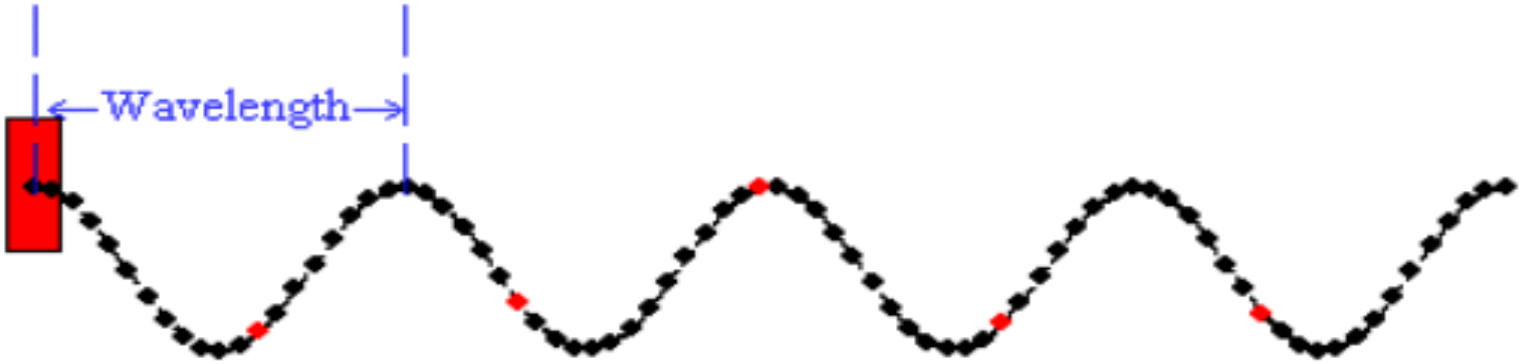
  to the motion of the wave.







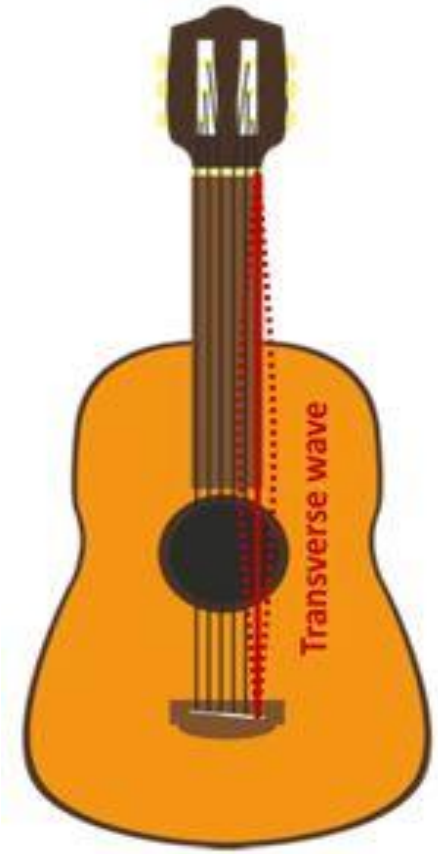
Transverse Wave

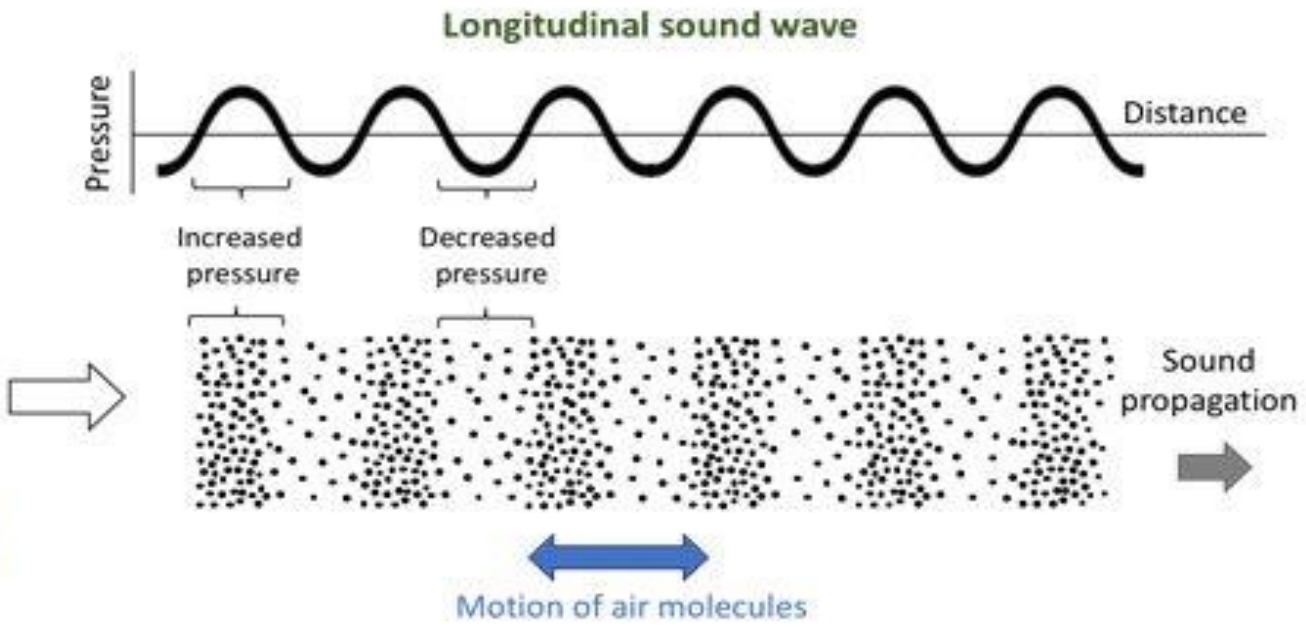


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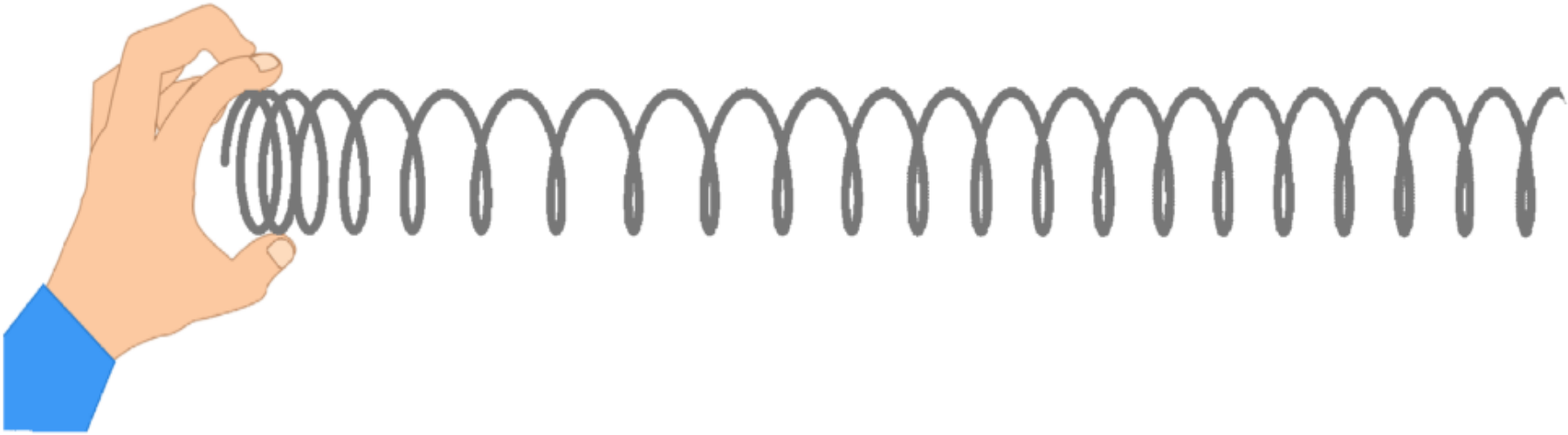
2- Longitudinal Waves

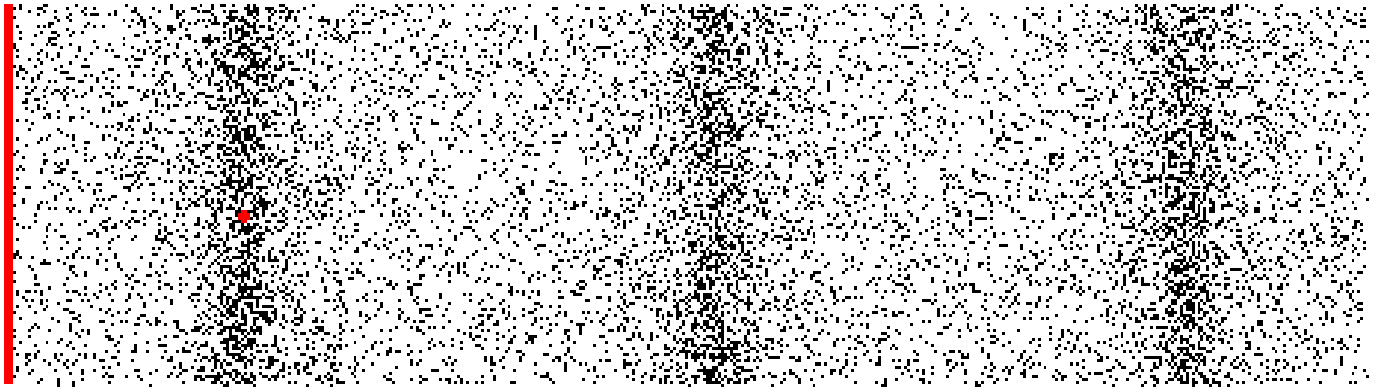
- Particles move **parallel** to the motion of the wave.





Particle Motion





©L.Perera

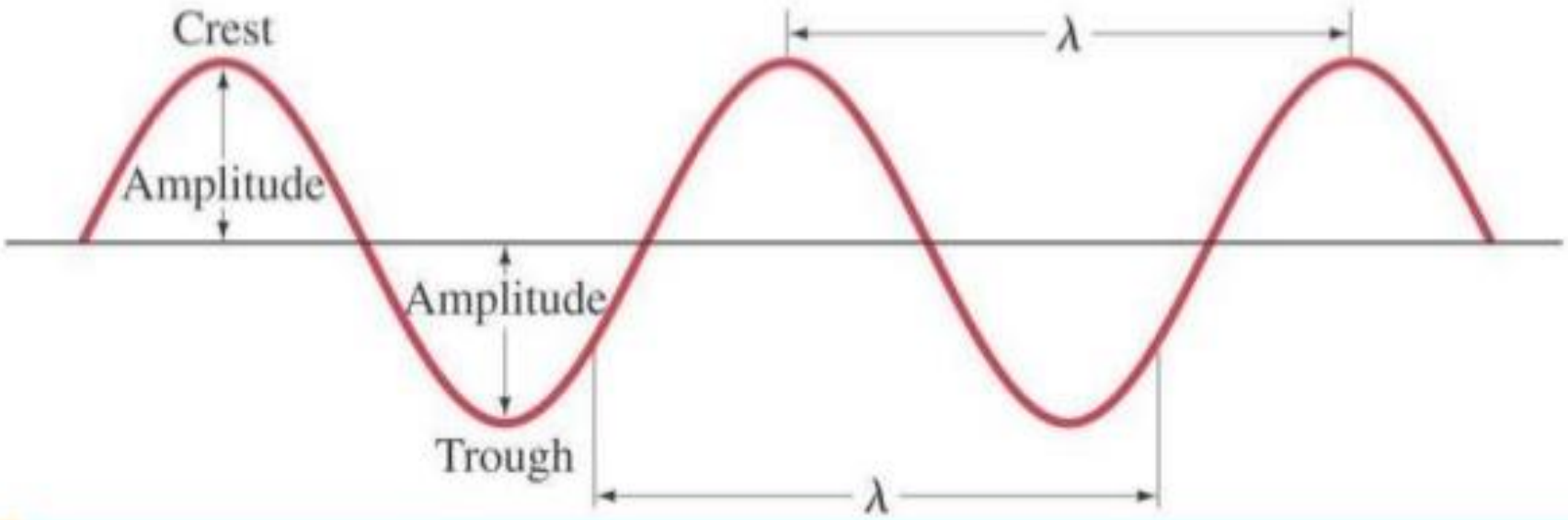
- **The transverse and longitudinal waves are called**

periodic waves

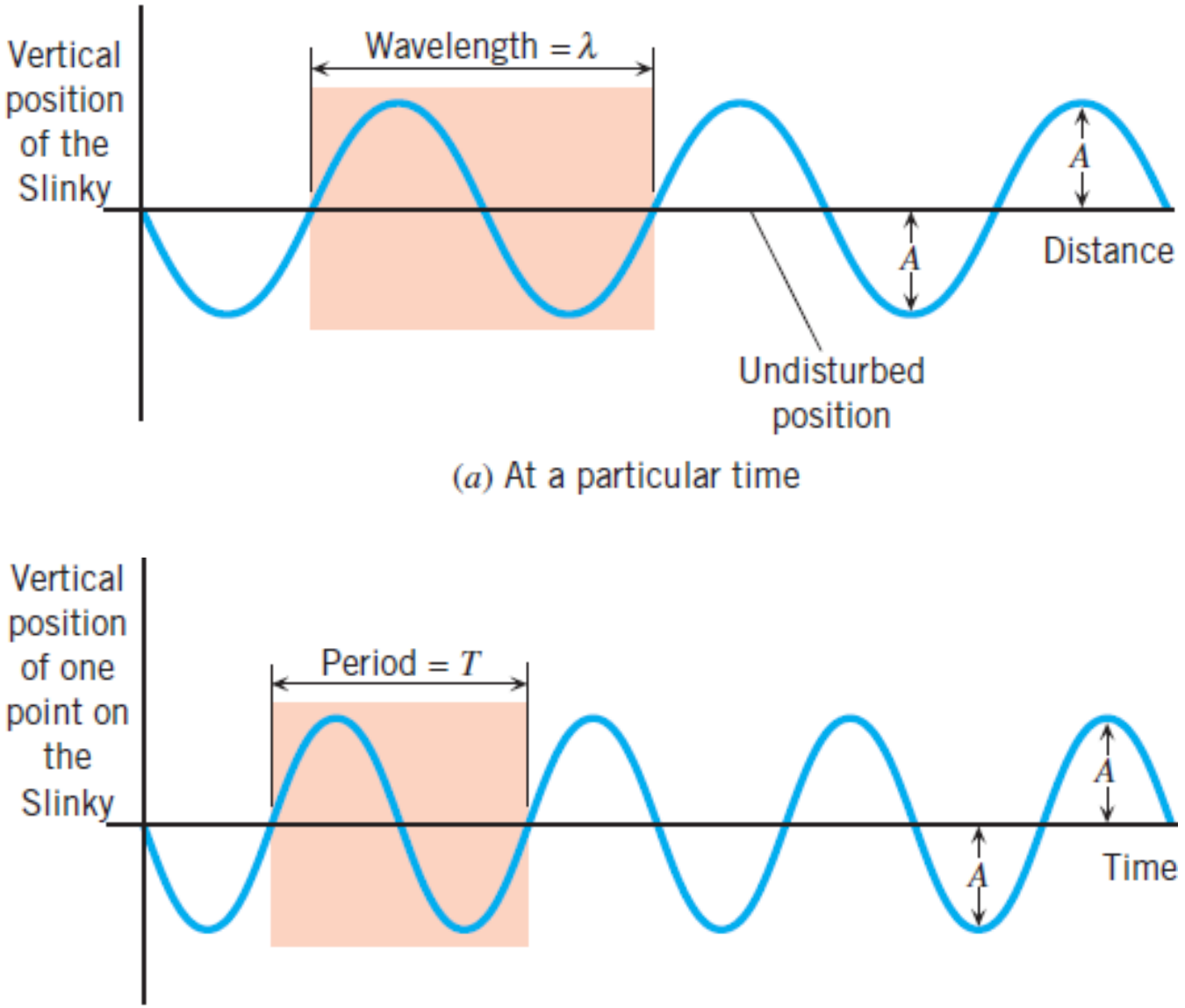
because they consist of cycles or patterns that are produced over and over again by the source.

Wave characteristics:

- **Amplitude,** A
- **Wavelength,** λ
- **Frequency,** f **and period,** T
- **Wave velocity,** $v = \lambda f$



- One *cycle* of a wave is shaded in color in both parts of the drawing.



❑ Wave Characteristics

1- *Frequency (f):*

The number of waves that passed a fixed point per second.

Unit: cycles per second, or hertz (Hz)

$$f = \frac{N}{t} = \frac{1}{T}$$

*2- Period (*Periodic time)* T :*

The time it takes a wave to travel a distance equal to a wavelength;

Unit: seconds

$$T = 1/f$$

3- Wave velocity (v):

Distance travelled by a wave crest in one period.

Unit: m/s

$$v = \lambda / T = f \lambda$$

4- The Angular Frequency

The Angular Frequency is how many rotational cycles per second.

(SI unit is rad/sec)

$$w = 2\pi f$$

5- Wave number

The wave number represent how many cycles are passing through a specific distance.

(SI unit is m^{-1})

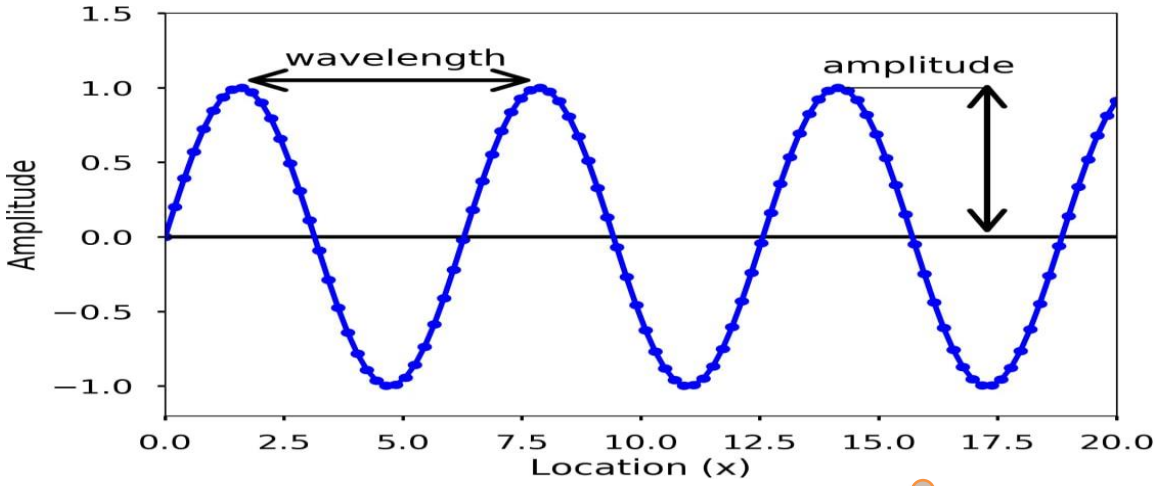
$$K = \frac{2\pi}{\lambda}$$

Wave equation as a function of location

$$y = A\sin(k\,x)$$

Wave number (k) (m^{-1})

$$k = \frac{2\pi}{\lambda}$$



Wave equation as a function of time

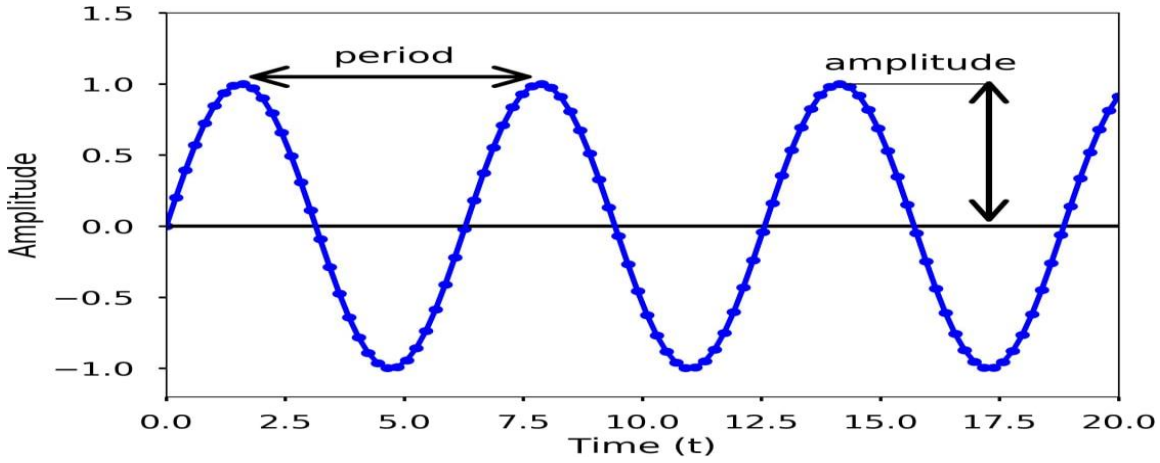
$$y = A\sin(wt)$$

Angular frequency (w) (rad/s)

$$w = 2\pi f$$

Speed of wave

$$v = \frac{w}{k} \quad or \quad v = \lambda f$$



General Wave equation

Wave move to right

$$y = A\sin(kx - wt)$$

Wave move to left

$$y = A\sin(kx + wt)$$

$$k = \frac{2\pi}{\lambda} \qquad w = 2\pi f \qquad v = \frac{w}{k}$$

Speed and acceleration of wave

Displacement (m)

$$y = A\sin(kx - wt) \longrightarrow y_{max} = A$$

Speed (m/s)

$$v = \frac{dy}{dt} = -Aw\cos(kx - wt) \longrightarrow v_{max} = -Aw$$

Acceleration (m/s^2)

$$a = \frac{dv}{dt} = -Aw^2\sin(kx - wt) \longrightarrow a_{max} = -Aw^2$$

Displacement (m)

$$y = A\sin(kx + wt) \longrightarrow y_{max} = A$$

Speed (m/s)

$$v = \frac{dy}{dt} = Aw\cos(kx + wt) \longrightarrow v_{max} = Aw$$

Acceleration (m/s^2)

$$a = \frac{dv}{dt} = -Aw^2\sin(kx + wt) \longrightarrow a_{max} = -Aw^2$$

Examples

A sinusoidal wave having the distance between two successive troughs of 2.00 m and amplitude 0.100 m travels on a string with a speed of 1.00 m/s to the right. For this wave, find

a) The frequency, The periodic time
b) The angular frequency
c) The wave number

$$\lambda = 2\,m \qquad A = 0.1m \qquad v = 1\ m/s$$

$$v = \lambda f \longrightarrow 1 = 2 \times f \longrightarrow f = 0.5Hz$$ a)

$$T = \frac{1}{f} = \frac{1}{0.5} = 2\,\mathrm{S}$$

$$w = 2\pi f = 2\pi \times 0.5 = \pi\, rad/s$$ b)

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{2} = \pi\, m^{-1}$$ c)

A transverse wave is observed to be moving along a lengthy rope. Adjacent crests are positioned 2.4 m apart. Exactly six crests are observed to move past a given point along the medium in 9.1 seconds. Determine the wavelength, frequency and speed of these waves.

$$\lambda = 2.4m \qquad N = 6 \qquad t = 9.1S$$

$$f = \frac{N}{t} = \frac{6}{9.1} = 0.66\,Hz$$

$$v = \lambda f = 2.4 \times 0.66 = 1.58\ m/s$$

Example

A transverse wave on a spring is described by the following equation

$$y = 0.12 \sin\left(\frac{\pi}{8}x + 4\pi t\right)$$

Determine the following:

a) The transverse wave speed and acceleration for an element on the string at t = 0.2 seconds at x= 1.6 m.
b) The wavelength of this wave
c) The periodic time
d) The speed of this wave propagation.

$$y = A\sin(kx + wt)$$

$$A = 0.12\,m \qquad w = 4\pi \qquad k = \frac{\pi}{8}$$

a) $$v = Aw\cos(kx + wt) = 0.12 \times 4\pi \times \cos\left(\left(\frac{\pi}{8} \times 1.6 + 4\pi \times 0.2\right) \times \frac{180}{\pi}\right) = -1.5 \text{ m/s}$$

$$a = -Aw^2 \sin(kx + wt) = -0.12 \times (4\pi)^2 \times \sin\left(\left(\frac{\pi}{8} \times 1.6 + 4\pi \times 0.2\right) \times \frac{180}{\pi}\right) = 0 \text{ m/s}^2$$

b) $$k = \frac{2\pi}{\lambda} \rightarrow \frac{\pi}{8} = \frac{2\pi}{\lambda} \longrightarrow \lambda = 16\,m$$

c) $$w = 2\pi f \rightarrow 4\pi = 2\pi f \longrightarrow f = 2\,Hz$$

$$T = \frac{1}{f} = \frac{1}{2} = 0.5 \text{ S}$$

d) $$v = \frac{w}{k} = \frac{4\pi}{\frac{\pi}{8}} = 32 \text{ m/s}$$

Thank You